Writing results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Writing results and limitations before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Writing the methods section

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Negative results and claim revision

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Writing results and limitations trace and record the first step you can explain without help.

Use W2: Trace the worked case for Writing results and limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Writing results and limitations explanation against the supplied facts before discussing it.

Use the worked case for Writing results and limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Candidate10/15, baseline9/15; no timings measured. Write result and two limits.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Writing results and limitations facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Writing results and limitations facts and keep your reasoning.

B passes9/12, A8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Writing results and limitations facts and keep your reasoning.

A report omits this qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

Explain how omitting it could change a reader's **interpretation**, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Writing results and limitations facts and keep your reasoning.

Candidate10/15, baseline9/15; no timings measured. Write result and two limits.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Writing results and limitations facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Writing results and limitations to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled writing results and limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing results and limitations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Writing results and limitations

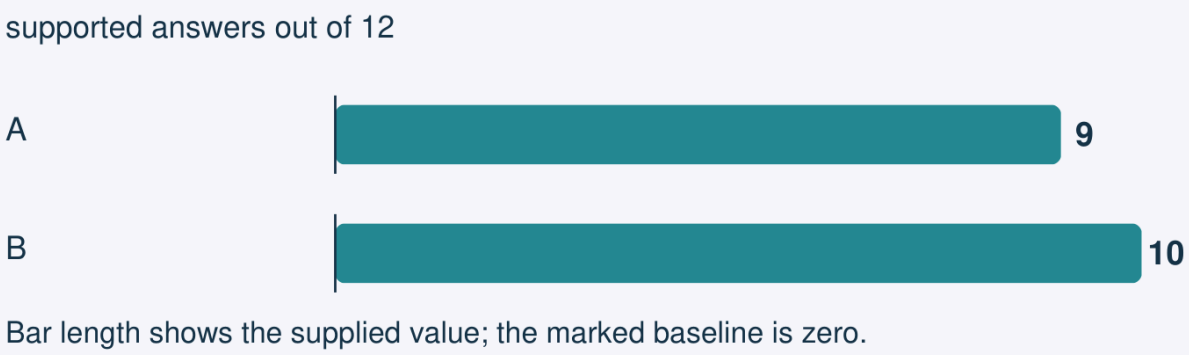
> Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection. Explain the labelled step before checking the changed condition.

Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.

A supports 9/12 answers; B supports 10/12. No user learning outcomes are recorded.

AI ACADEMY / AI-7/31

Supported answers in one collection



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/31

Writing results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Report the observation

A is 75%; B is about 83.3%, a one-case improvement.



2. Link to the method

Name the collection, rubric and twelve-case denominator.



3. State limits

Do not infer that children learn more from this answer-support measure.

CHANGE ONE THING

Write "B makes every student smarter".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compute the supported-answer rates for 9/12 and 10/12. What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Do not infer that children learn more from this answer-support measure.

1. Report the **observation**

2. Link to the method

3. State limits

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State limits Do not infer that children learn more from this answer-support measure.

Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Write "B makes every student smarter". Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Remove that sentence as unsupported by the study.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

Repair a missing building block

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

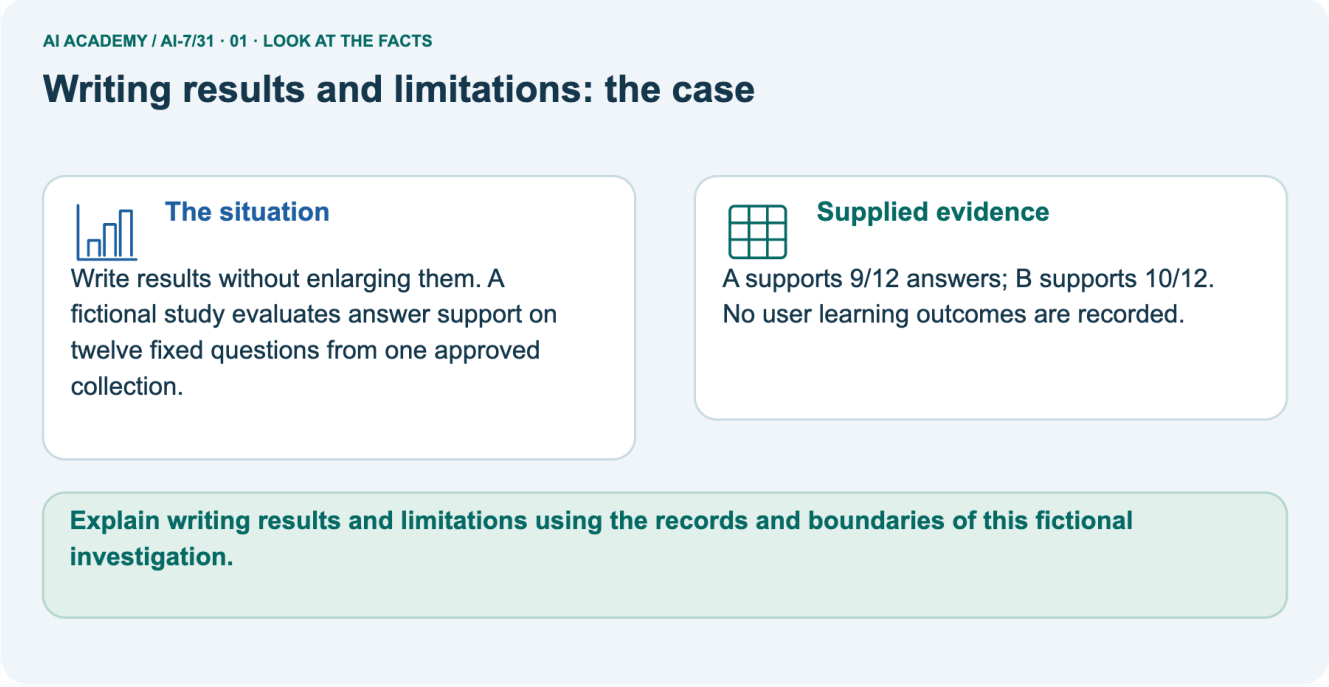
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Writing results and limitations picture from its labels, then record a relevant limit.

01. Writing results and limitations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

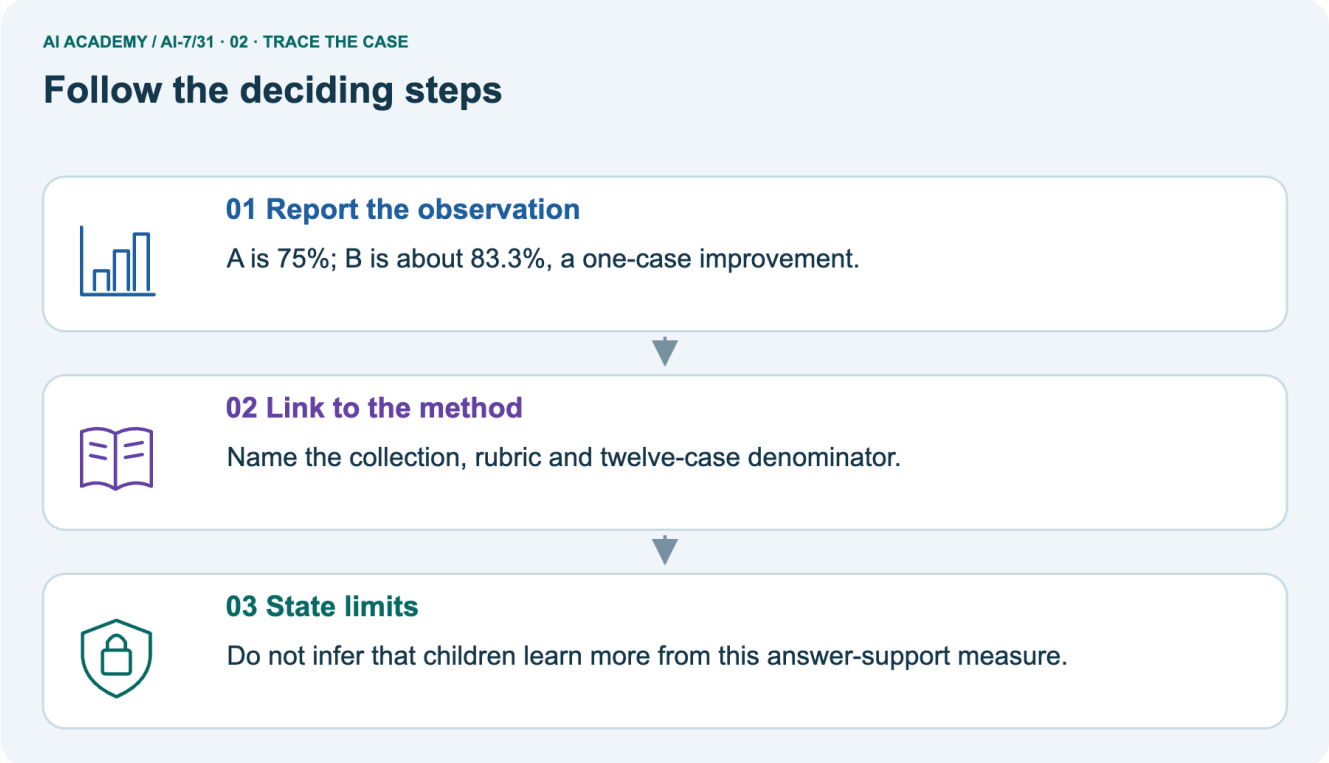
Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection. Supplied facts: A supports 9/12 answers; B supports 10/12.

No user learning outcomes are recorded.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



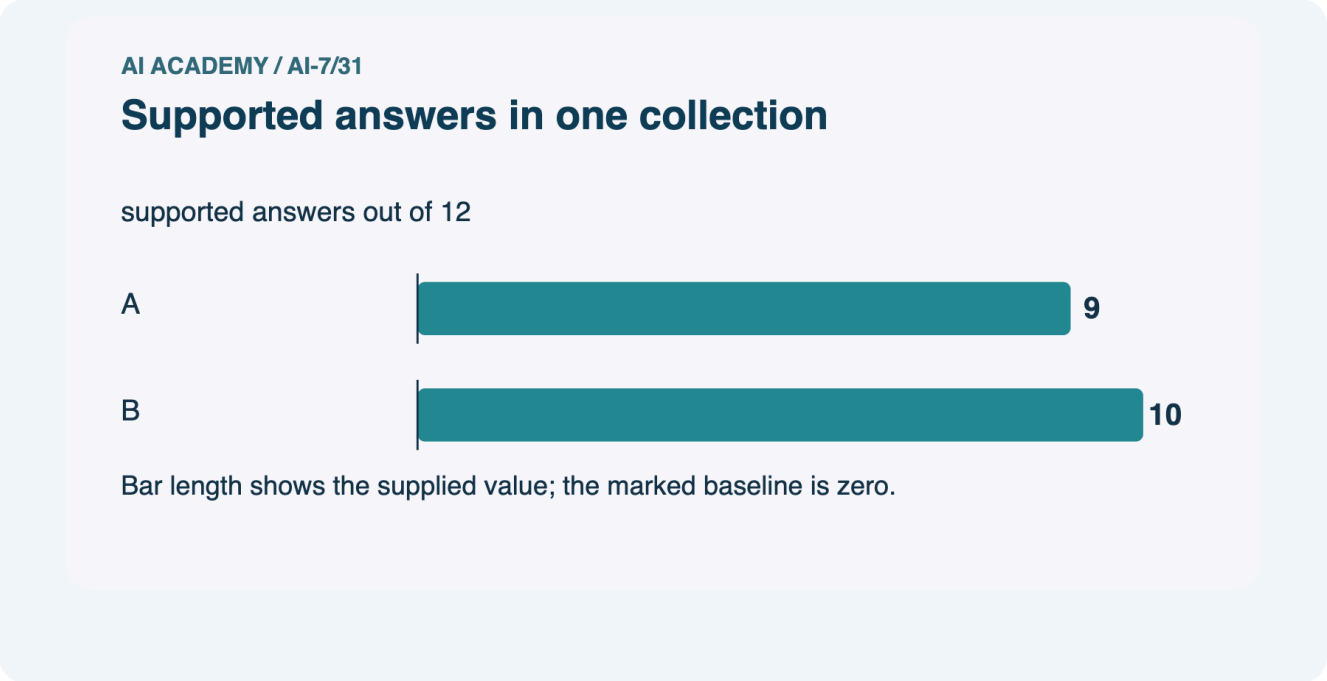
Follow the labelled picture with the matching explanation. Open the vector diagram

Report the **observation**: A is 75%; B is about 83.3%, a one-case improvement. → Link to the method: Name the collection, rubric and twelve-case denominator. → State limits: Do not infer that children learn more from this answer-support measure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Supported answers in one collection



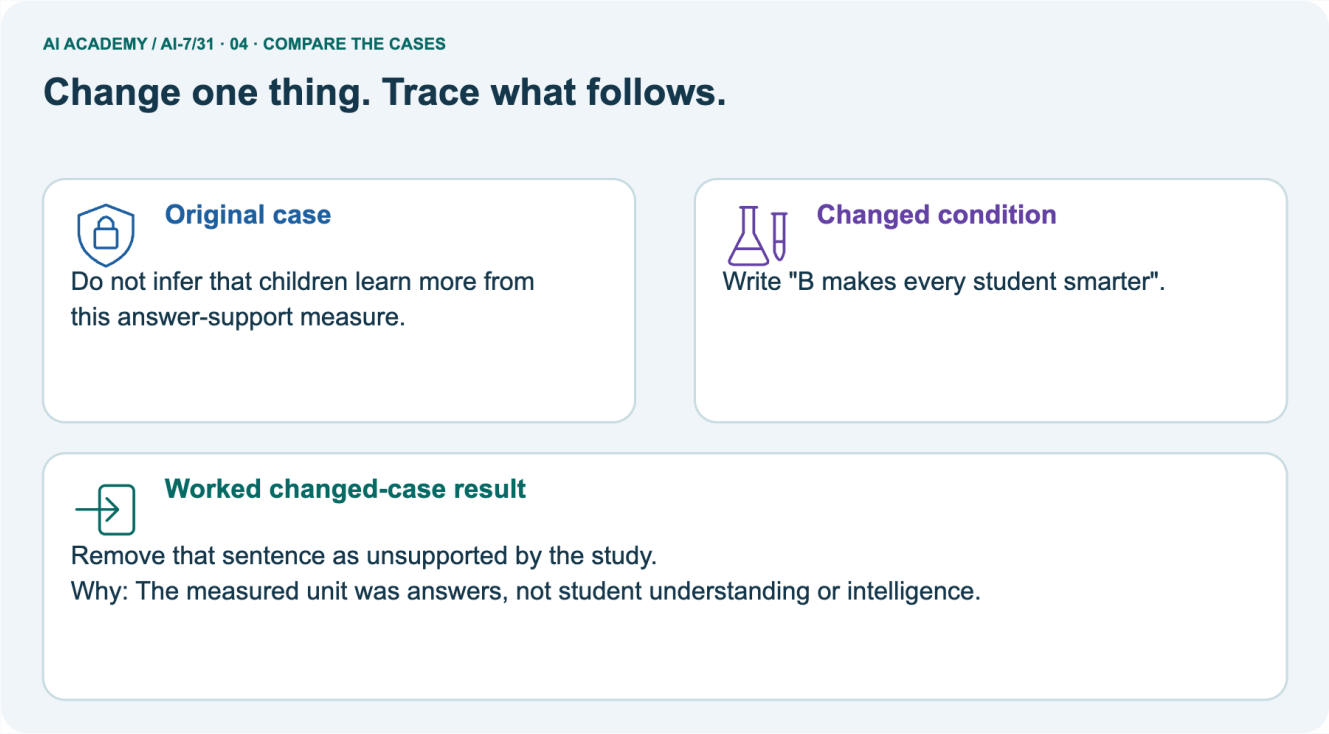
Follow the labelled picture with the matching explanation. Open the vector diagram

Supported answers in one collection. Units: supported answers out of 12. A: 9. B: 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

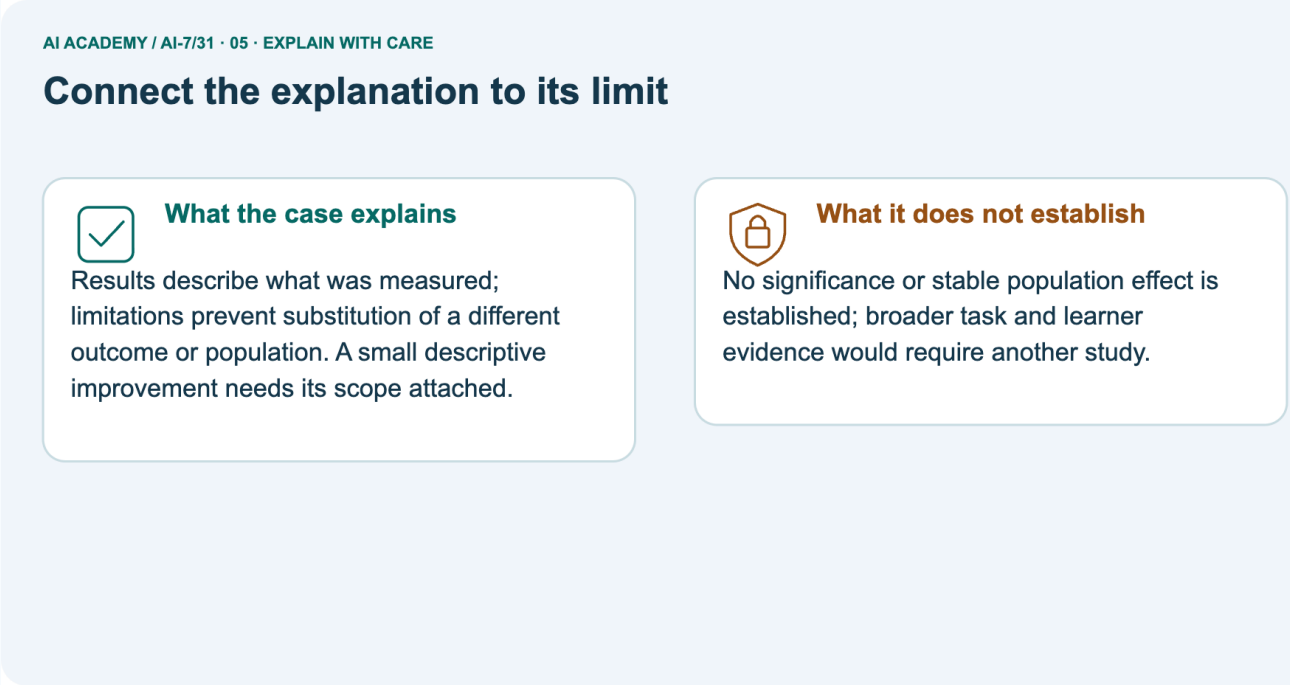
Original case: Do not infer that children learn more from this answer-support measure. Changed condition: Write "B makes every student smarter". Expected result: Remove that sentence as unsupported by the study.

Reason: The measured unit was answers, not student understanding or intelligence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

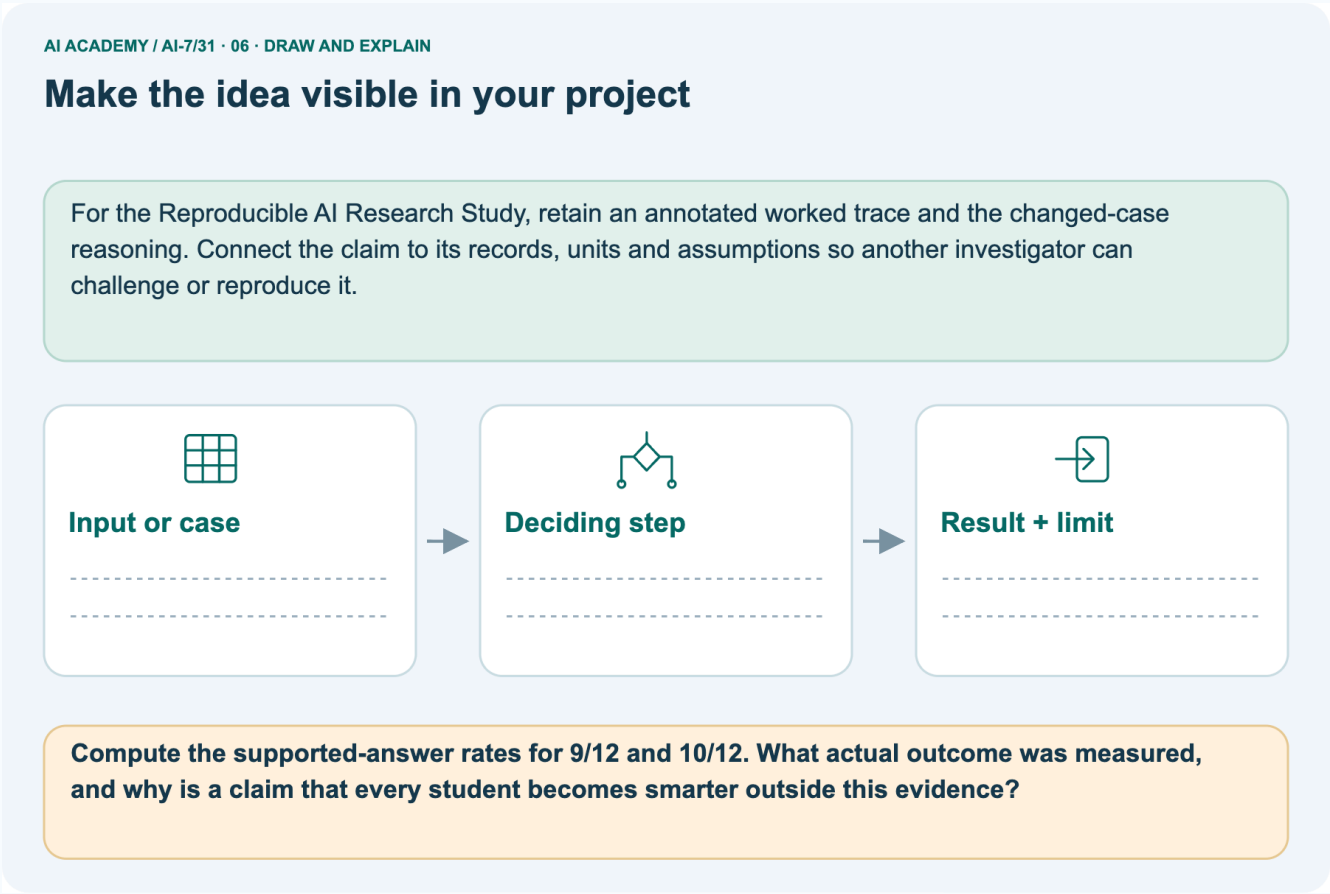
Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

Limit: No significance or stable population effect is established; broader task and learner **evidence** would require another study.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.